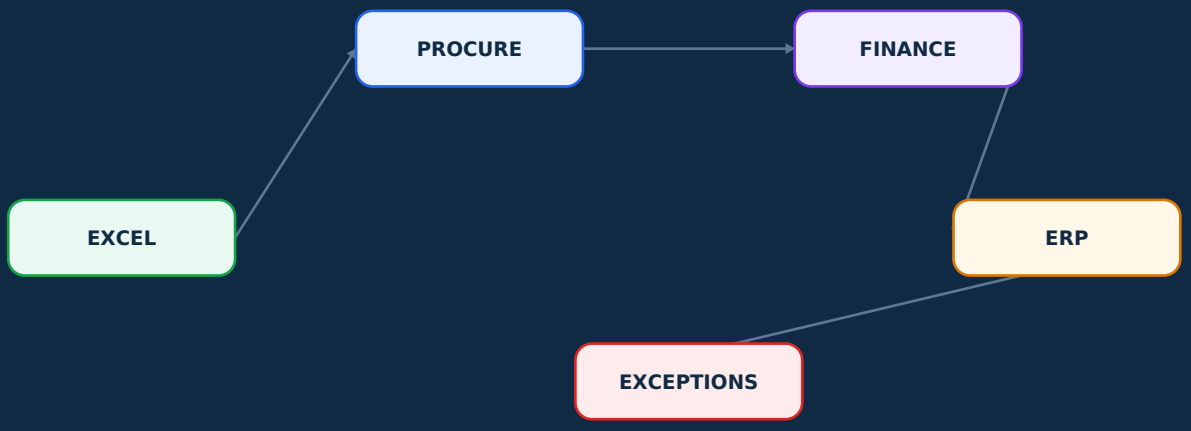
TECHNICAL BLUEPRINT

vProcly

Procurement, Finance and ExceptionOS

On-premises architecture with Excel-first ingestion



Architecture edition 1.0 | September 2026

Portable • Auditable • Scalable

Executive summary

Architecture recommendation

Build vProcly as a portable procurement-to-finance control platform. Procurement remains the operational entry point; finance is added through budgets, commitments, invoice control, payment readiness and cash visibility. The customer's ERP or accounting application remains the statutory system of record.

What the platform becomes

1 Procurement workspace

Requests, approvals, sourcing, RFQs, quotation comparison, award and purchase-order coordination.

2 Finance control workspace

Budget checks, commitments, accrual visibility, invoice matching, payment readiness and cash forecasting.

3 ExceptionOS

Continuously detects delays, commercial leakage, compliance breaches, mismatches and financial exposure.

4 Integration and evidence layer

Excel, API, email and ERP inputs are normalized into one canonical model with provenance and audit history.

Important boundary

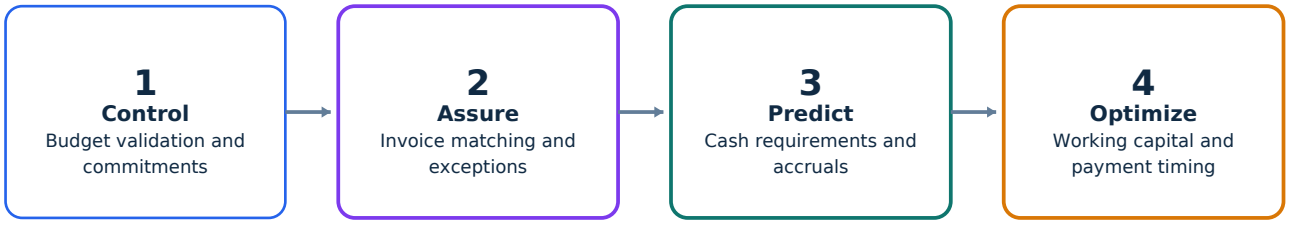
Do not initially build a general ledger, payroll, banking core, tax engine or full accounting suite. Integrate with those systems. vProcly should own procurement-related financial control, not statutory accounting.

Product boundary: what vProcly owns

A clean ownership boundary prevents vProcly from becoming another large ERP. It should be the system of action and intelligence around a purchase, while the ERP remains the accounting system of record.

vProcly owns	Shared / synchronized	ERP or finance system owns
Purchase requests and intake	Supplier and item references	General ledger
Approvals and sourcing	Purchase orders	Statutory books
Quotes and award evidence	Goods receipts	Tax filing
Supplier commitments	Invoices and payment status	Bank posting
Budget availability checks	Cost centres and projects	Treasury execution
Exceptions and resolution	Exchange rates and tax codes	Final financial statements
Payment-readiness decision	Vendor balances	Payment release

Finance expansion sequence

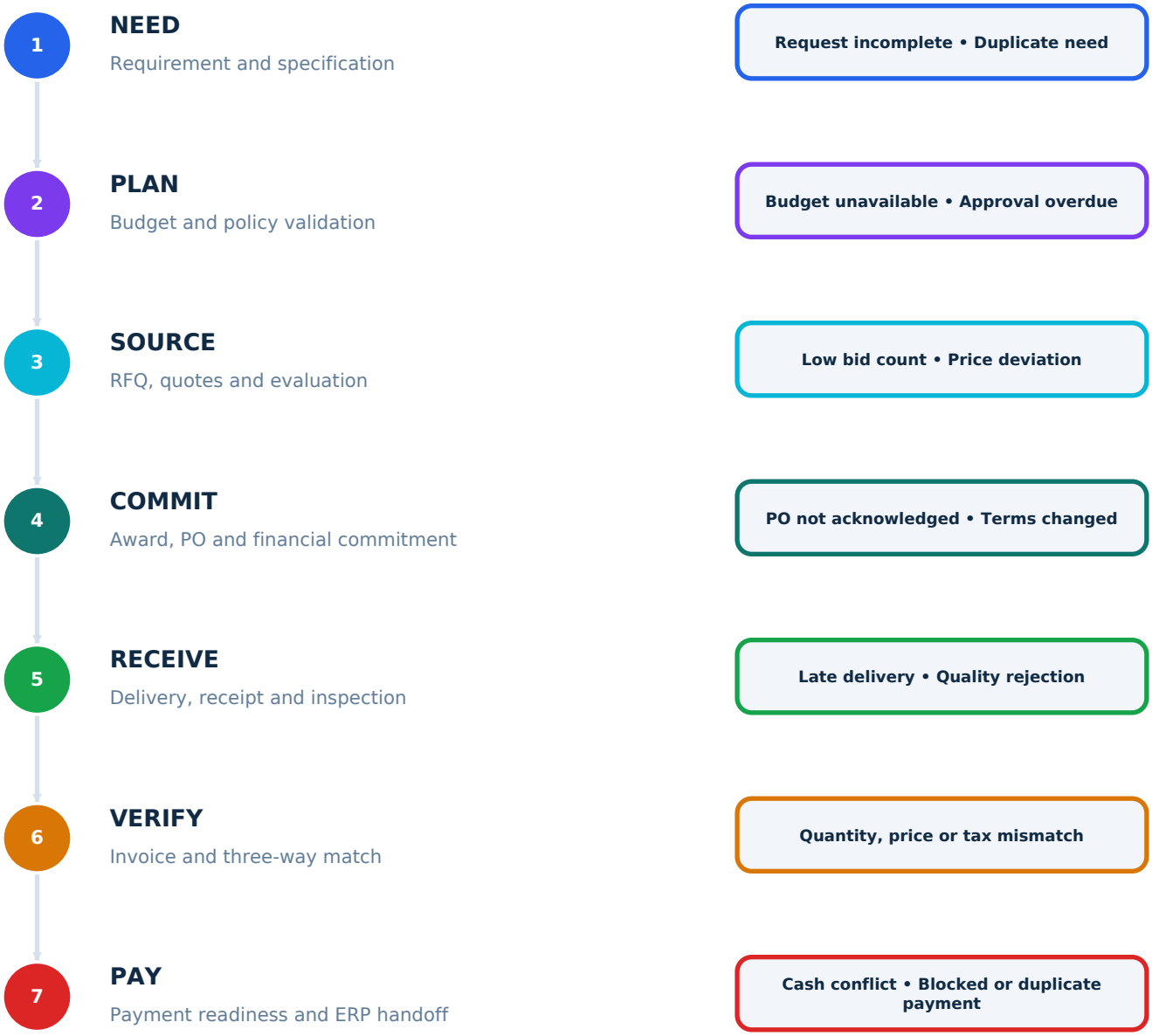


Positioning

vProcly manages every purchase and its financial consequences, while ExceptionOS prevents operational and financial leakage.

End-to-end procurement-to-finance lifecycle

Every purchase is one continuously monitored case. Finance is not a disconnected module added after procurement; it begins when money is requested and continues through commitment, accrual, invoice and payment readiness.



Capability map

The product is divided into vertical business capabilities and horizontal platform capabilities. ExceptionOS, evidence and audit operate across every stage.

INTAKE

Excel, email, forms, API

SOURCE

RFQ, quotes, negotiation

BUY

Approval, award, PO

FULFIL

Commitments, receipt

FINANCE

Budget, accrual, match

SUPPLIER

Onboarding, risk, score

EXCEPTIONS

Detect, prioritize, resolve

ANALYTICS

Spend, cycle, cash, KPIs

Horizontal foundations

WORKFLOW
Temporal

EVENTS
NATS JetStream

IDENTITY
Keycloak

DATA
PostgreSQL

FILES
MinIO / S3

AI
Model gateway

Design rule

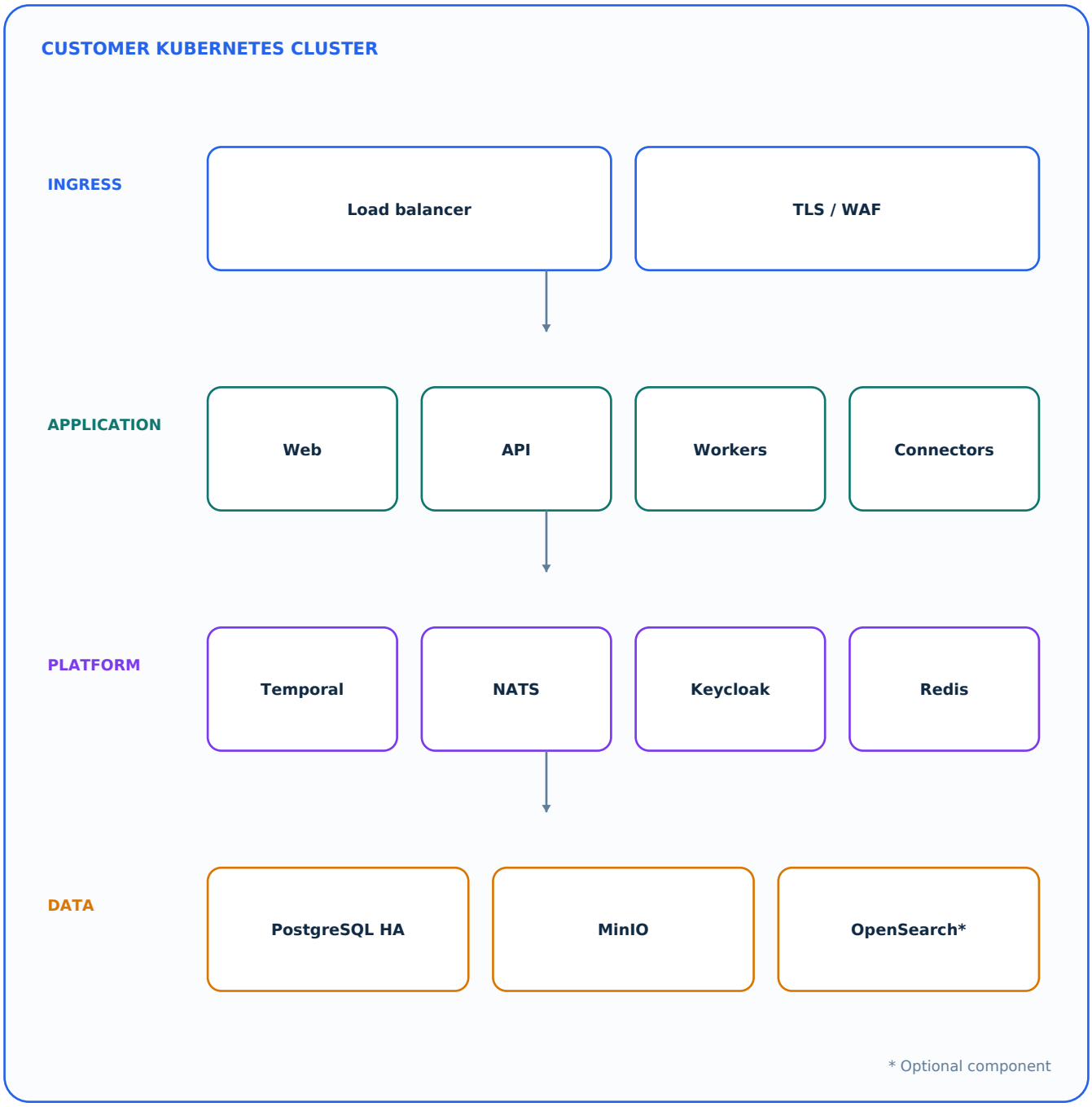
All channels produce the same canonical objects. An Excel-imported invoice, an API invoice and an ERP-synchronized invoice are processed by the same matching rules and exception detectors.

Logical system architecture



On-premises deployment topology

The platform ships as signed OCI containers and Helm charts. No public-cloud service is mandatory. Customers can connect their own load balancer, identity provider, storage and monitoring systems.



Deployment profiles and sizing

Profile	Typical use	Topology	Indicative starting capacity
Developer	Local development and demos	Docker Compose	8 CPU, 16 GB RAM
Small production	One company or plant	Single node or 3-node k3s/RKE2	16-24 CPU, 32-64 GB RAM
Enterprise HA	Multiple plants and integrations	3+ Kubernetes nodes; HA data services	3 nodes × 16 CPU, 64 GB RAM
Large enterprise	High import and analytics volume	Separate compute, data and worker pools	Sized from measured workload

Scaling model

API

Scale replicas by CPU, latency and request rate

Excel

Scale workers by queue depth and workbook size

Documents

Separate OCR and extraction worker pools

Connectors

Isolate each ERP/customer connector

Analytics

Add read replica, then ClickHouse

Database

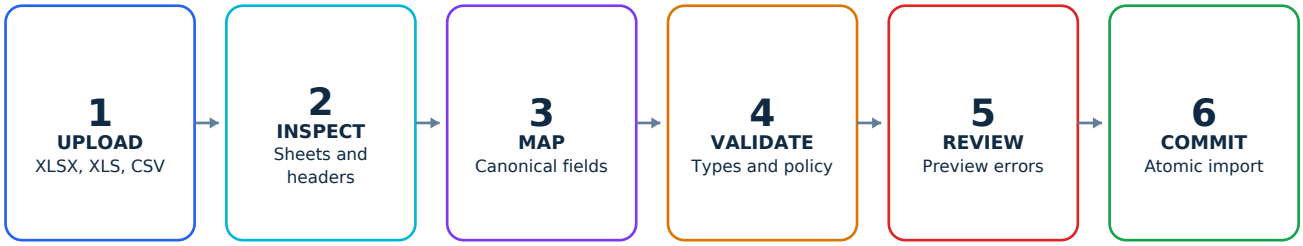
PostgreSQL HA, partitioning, tenant isolation

Sizing principle

Publish tested baseline profiles, but size production from actual concurrency, monthly line volume, workbook size, document pages, integration frequency, retention period and recovery objectives.

Excel-first procurement ingestion

Excel is treated as a governed integration channel. Users can repeatedly import their existing operational files without losing validation, traceability or control.



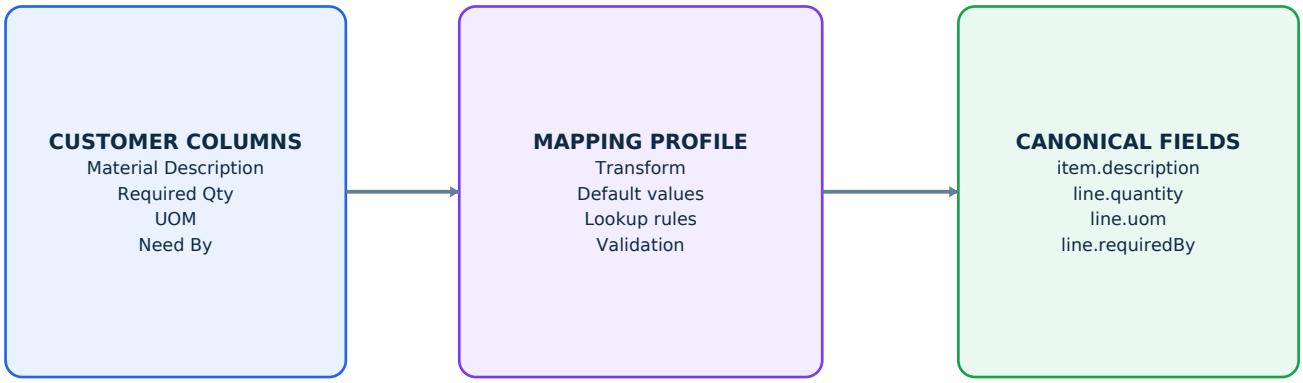
Import type	Examples	Result
Master data	Items, suppliers, UOM, projects	Validated canonical master records
Procurement	Indents, BOM, BOQ, RFQ lines	Requests and sourcing cases
Commercial	Supplier quote sheets, price lists	Normalized quotations
Fulfilment	Open POs, delivery schedules, GRNs	Commitments and receipts
Finance	Budgets, invoices, payment status	Controls, matching and projections
History	PO and spend history	Benchmarks and supplier performance

Non-negotiable

No Excel worker writes directly into live procurement tables. It writes normalized rows to staging; a user reviews the result; the core application commits valid records and emits normal domain events.

Excel mapping studio

The mapping studio converts customer-specific spreadsheets into reusable, versioned import profiles. AI may suggest mappings, but users approve them.



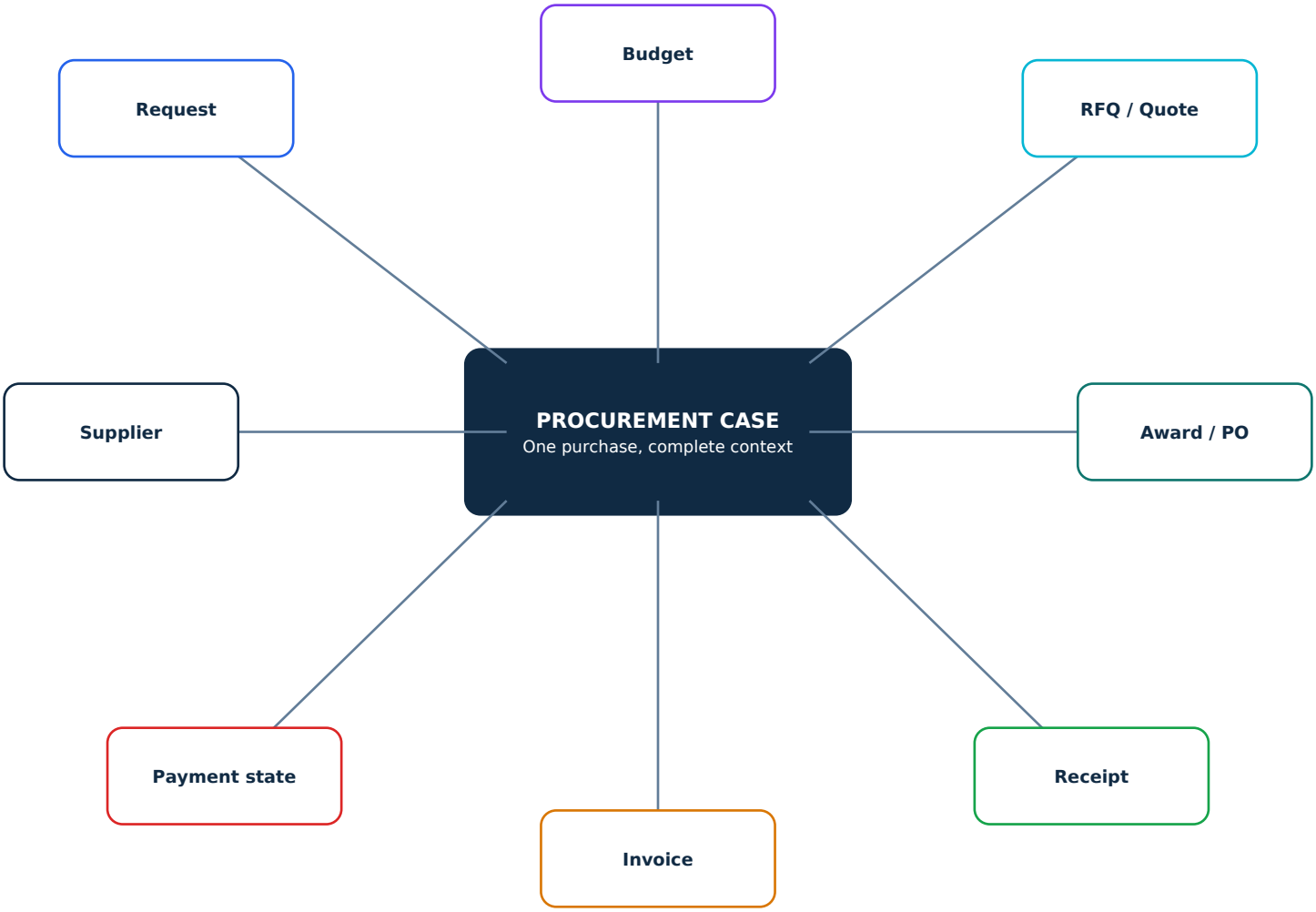
Capability	How it works
Template detection	Uses workbook fingerprint, headers and past profiles to select a mapping.
Header discovery	Finds the true header even when title rows, merged cells or notes appear first.
Transformations	Dates, decimals, currency, UOM, codes, text cleanup and lookup replacement.
Defaults	Adds tenant, plant, currency, cost centre or request type where omitted.
Preview	Shows original value, transformed value, warning and target field side by side.
Versioning	Keeps prior mappings so an older workbook format remains reproducible.
Drift detection	Warns when columns, sheet names or value patterns change unexpectedly.

Robust processing

Use an isolated Python ingestion service with openpyxl/calamine and a sandboxed LibreOffice worker for formula recalculation. Never execute Excel macros.

Canonical procurement and finance model

A shared canonical model is the key to flexibility. Excel, ERP, APIs, forms and email all map into the same objects, so workflows and exceptions do not contain channel-specific logic.



Evidence attached to every object

Source system, external ID, document, cell or page location, import batch, event correlation ID, actor, timestamp, confidence, validation status and version.

Finance integration: the control model

Finance enters the lifecycle through five controlled objects. These give the CFO visibility without requiring vProcly to become the statutory ledger.

1 BUDGET

Authorized spending capacity for cost centre, project, category and period.

2 RESERVATION

Temporary budget hold created when a request reaches approval.

3 COMMITMENT

Expected financial obligation created from an approved award or PO.

4 ACCRUAL SIGNAL

Estimated liability when goods/services are received before invoicing.

5 PAYMENT READINESS

Verified status indicating that invoice conditions are satisfied.

Accounting handoff

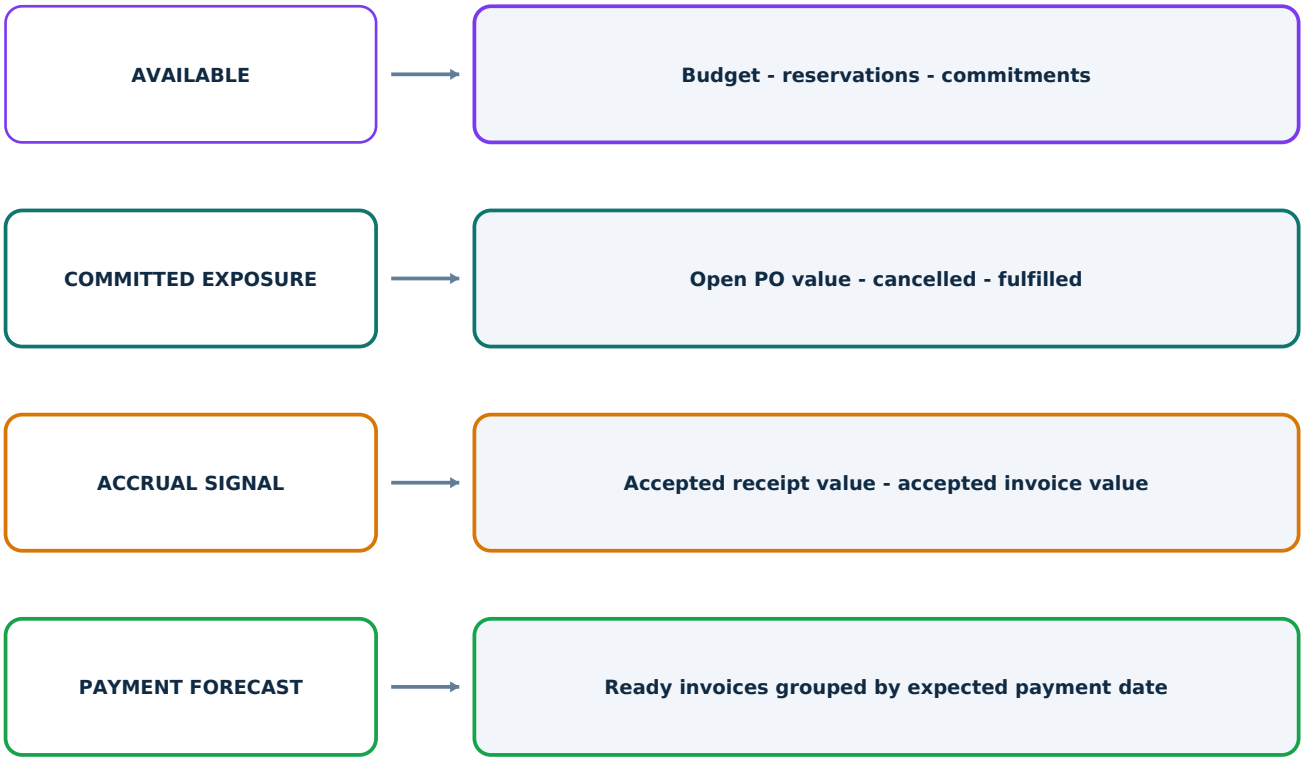
vProcly sends verified documents, match results, approval evidence and payment-readiness status to the ERP. The ERP creates accounting entries, tax postings and payment transactions.

How money changes through one purchase

The platform maintains a financial shadow state for operational control. This state is reconciled with the ERP but never silently replaces official accounting balances.

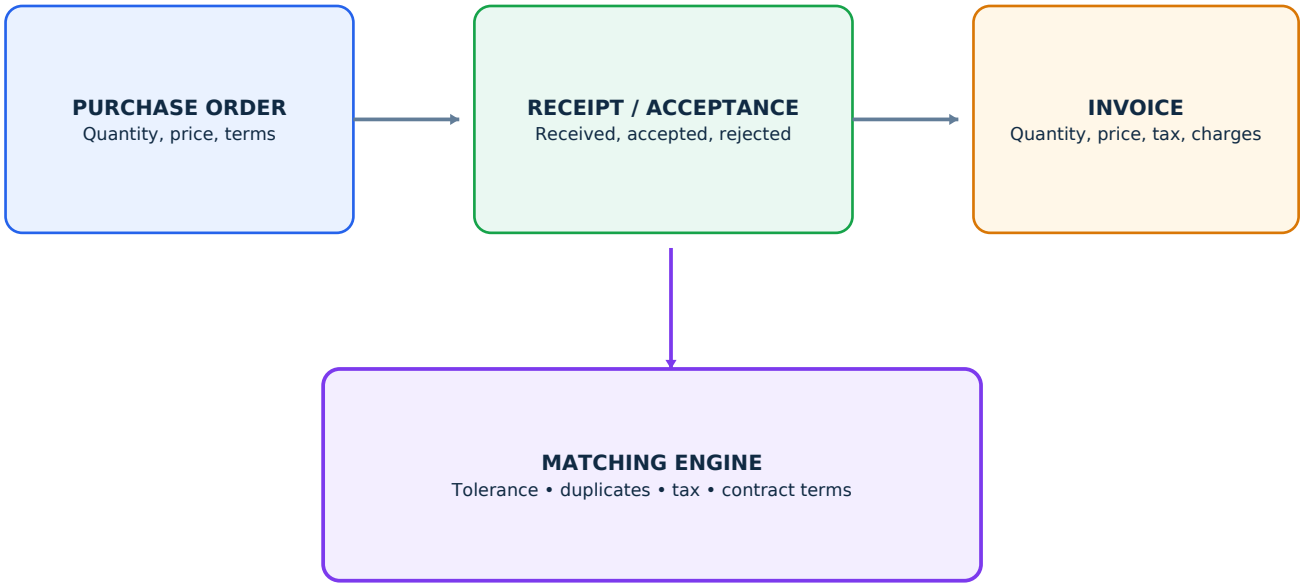
Moment	Budget	Reserved	Committed	Accrued	Invoiced
Budget loaded	₹10,00,000	₹0	₹0	₹0	₹0
Request submitted	₹10,00,000	₹3,00,000	₹0	₹0	₹0
PO issued	₹10,00,000	₹0	₹2,85,000	₹0	₹0
Goods received	₹10,00,000	₹0	₹2,85,000	₹2,85,000	₹0
Invoice accepted	₹10,00,000	₹0	₹2,85,000	₹0	₹2,85,000
ERP reports payment	₹10,00,000	₹0	₹0	₹0	₹0

Control equations



Invoice assurance and payment readiness

Payment readiness is an operational assurance result, not the payment itself. It packages evidence for finance and the ERP.



Result	Meaning	Next action
Matched	All required evidence is present and within tolerance	Mark payment ready
Minor variance	Difference is within configured review threshold	Route for controlled approval
Blocked	Material price, quantity, tax or supplier issue	Raise finance exception
Duplicate risk	Invoice number, amount or fingerprint resembles prior invoice	Freeze and investigate
Missing evidence	Receipt, inspection or service acceptance unavailable	Request evidence

Separation of duties

The person resolving an invoice exception should not automatically receive permission to release payment. High-risk overrides require independent approval and complete audit evidence.

ExceptionOS across procurement and finance



Procurement exception	Finance exception
Incomplete request	Budget unavailable or expired
Insufficient supplier responses	Commitment exceeds approved amount
Price above historical range	Invoice price or tax mismatch
PO not acknowledged	Duplicate invoice risk
Delivery at risk	Accrual missing or inaccurate
Quality rejection	Payment-ready invoice approaching due date
Supplier compliance expired	Bank detail changed unexpectedly

Priority model

Priority combines financial impact, operational criticality, probability, urgency and compliance risk. Each component is stored and displayed so users understand why an exception is ranked highly.

Event-driven execution

Core transactions emit versioned domain events using a transactional outbox. Events activate workflows, projections and exception detectors without tightly coupling modules.



Reliability rules

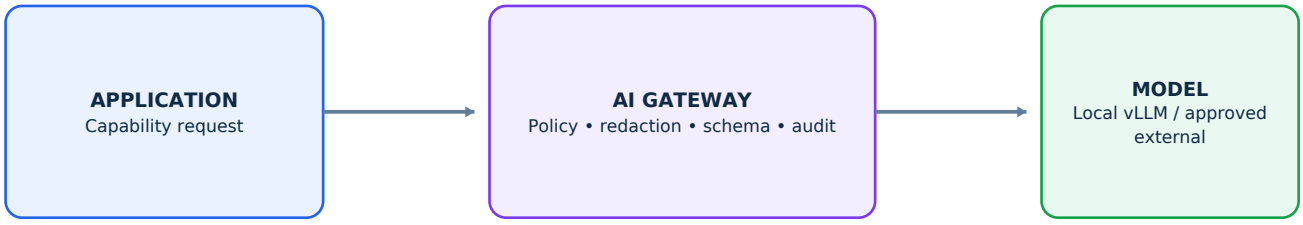
Use outbox publishing, idempotent consumers, retry with exponential backoff, dead-letter queues, correlation IDs and reconciliation jobs. Never assume that a message is delivered exactly once.

AI architecture and safety boundary

AI improves interpretation and user productivity. Deterministic rules continue to control money, approvals, authorization and final transactions.

AI may assist	Rules must control	Human or ERP must authorize
Map Excel columns	Budget availability	Final supplier award
Extract quotation fields	Approval thresholds	Policy override
Normalize descriptions	Invoice tolerance	Bank-detail acceptance
Summarize deviations	Duplicate checks	PO issuance
Draft supplier messages	Payment-readiness rules	Payment release
Recommend resolution	Role permissions	Statutory posting

On-prem model gateway

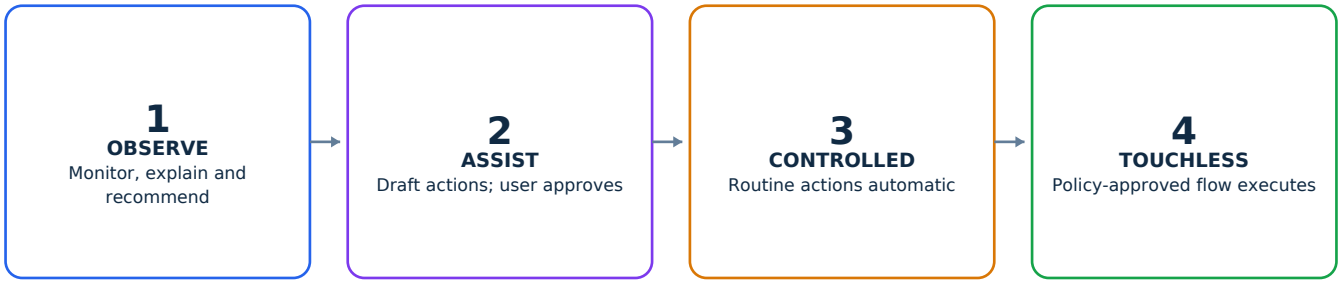


Graceful degradation

Procurement, finance controls, matching and workflows must remain operational when AI is disabled or temporarily unavailable.

Human-in-the-loop and touchless operation

vProcly supports assisted and autonomous procurement in the same platform. Policy, risk and confidence determine whether the system executes an action or routes it to a person.



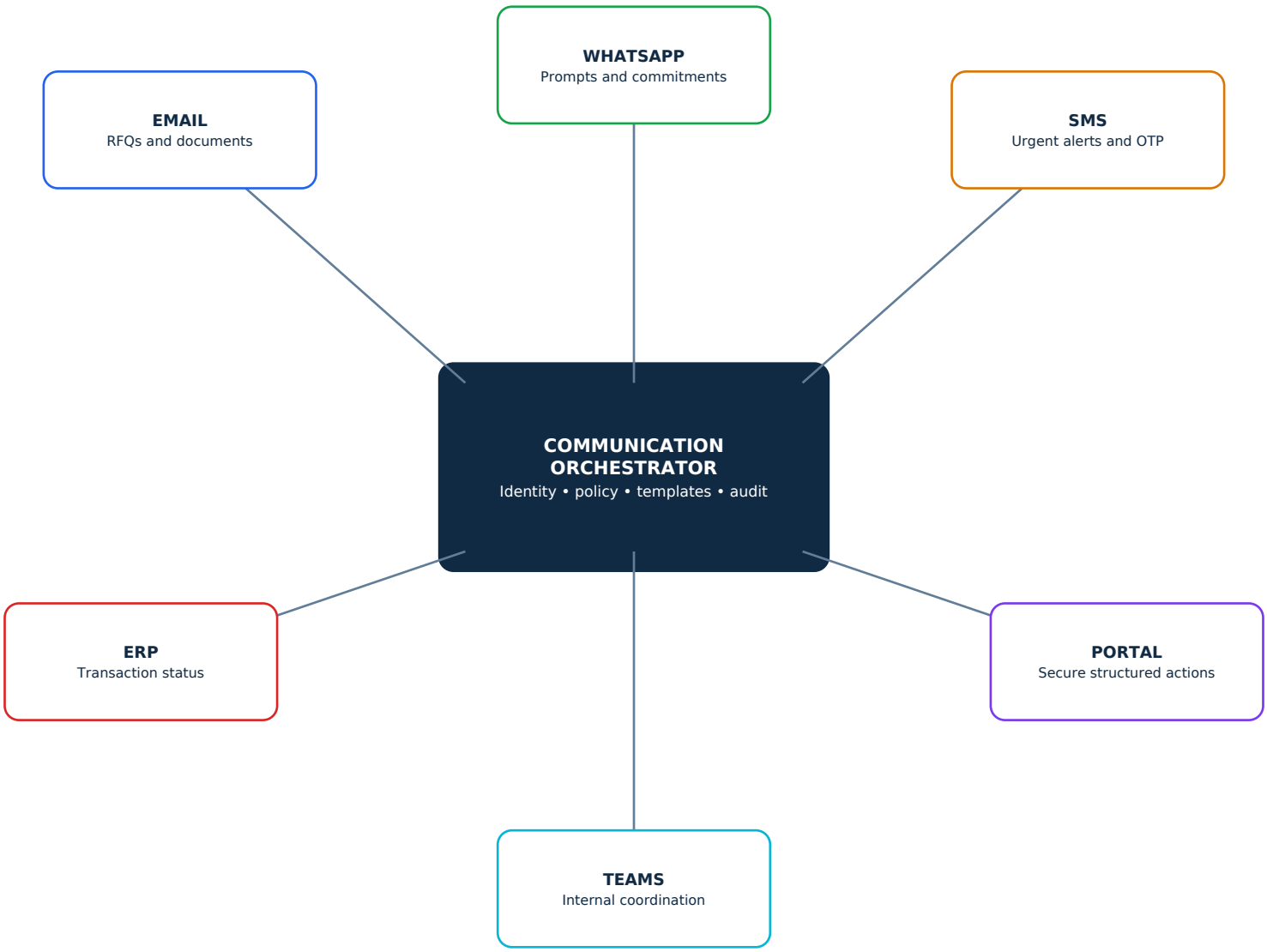
Decision factor	More autonomy	More human control
Purchase value	Low and within threshold	High or unusual
Supplier	Approved and contracted	New, blocked or high-risk
Specification	Standard catalogue item	Custom or technically critical
Price	At contract / within tolerance	Material deviation
Budget	Available and reserved	Unavailable or near limit
Evidence confidence	Complete and high-confidence	Missing or ambiguous
Exception state	No open exception	Active material exception

Operating principle

Autonomous by policy, human by exception. Touchless does not mean uncontrolled: every automatic action still requires an authorized rule, verified identity, evidence and audit history.

Omnichannel procurement communication

Email, WhatsApp, SMS and the supplier portal are adapters around one governed conversation timeline. The workflow never contains channel-specific business logic.



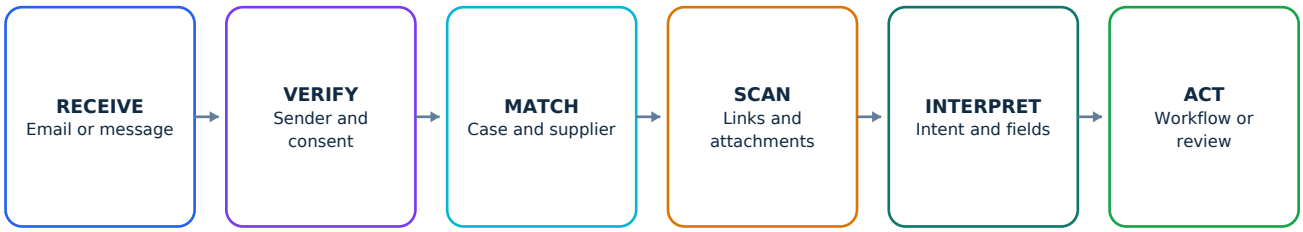
Unified conversation record

Every inbound or outbound message is linked to the tenant, procurement case, supplier, RFQ or PO, exception, recipients, attachments, delivery state, extracted intent and approval status.

Channel responsibilities and safeguards

Channel	Best suited for	Do not use for
Email	RFQs, quotes, documents, detailed clarification and threaded replies	Unverified high-risk approvals
WhatsApp	Reminders, acknowledgement, delivery dates, dispatch proof and secure links	Sensitive bank data or complete contracts
SMS	OTP, urgent alerts, due-date reminders and short secure links	Quotations, negotiation or complex decisions
Supplier portal	Structured quotes, compliance files, PO acceptance and invoice upload	Casual conversations
Teams / internal chat	Approver notification, collaboration and exception alerts	Unauthenticated supplier commitments

Inbound message processing



Uncertain identity

If the sender, tenant, procurement case or intent cannot be identified confidently, the message is quarantined for human review. It must never update financial or supplier records speculatively.

Secure links

Messages should contain minimal data and an expiring authenticated link for approvals, quotations, contracts, bank changes and other sensitive actions.

Action permissions and mandatory human gates

Each executable action receives an autonomy classification. The classification can vary by tenant, category, value, business unit and exception severity.

Action	Default mode	Reason
RFQ deadline reminder	Automatic	Low-risk operational communication
Request missing quotation information	Automatic	Clarification without commercial commitment
Ask for dispatch evidence	Automatic	Evidence collection
Draft negotiation message	Human-approved	Could change commercial terms
Select final supplier	Human decision	Material financial and operational choice
Issue high-value PO	Human approval	Creates financial commitment
Accept supplier bank change	Independent verification	Fraud and payment-diversion risk
Override invoice mismatch	Segregated finance approval	Changes payment eligibility
Release payment	ERP / authorized finance user	Banking and statutory control

Secure approval path



Approval integrity

A casual 'OK' reply should not authorize a material purchase unless the customer's explicit policy permits it and sender identity, action scope, context and evidence are strongly verified.

Touchless procurement with exception handoff

A low-risk catalogue purchase can run end to end without manual handling. Any failed condition pauses the workflow and creates an explained human task.

Touchless eligibility condition	Required state
Category and item	Approved catalogue or contract item
Value	Within autonomous threshold
Supplier	Approved, compliant and not blocked
Budget	Available and successfully reserved
Price	Contracted or within configured tolerance
Evidence	Complete and above confidence threshold
Exceptions	No material open exception

Automatic lifecycle



Mandatory pause condition	Human task created
Low extraction confidence	Verify source information
Budget unavailable	Choose funding or reject
New or risky supplier	Complete onboarding and review
Price/specification deviation	Commercial or technical decision
Late delivery or quality rejection	Choose recovery action
PO-receipt-invoice mismatch	Finance exception review
Bank-detail change	Independent supplier verification

Outcome

The user does not monitor every step. vProcly completes permitted routine work and brings a person in only when judgment, authority or independent verification is required.

Security, audit and on-prem controls

IDENTITY

Keycloak, Active Directory, SAML/OIDC, MFA and service accounts.

ACCESS

Tenant, legal entity, plant, category, role and object-level authorization.

DATA

TLS, encryption at rest, secrets vault and short-lived file links.

NETWORK

Kubernetes network policies and outbound allowlists for connectors.

FILES

Quarantine, MIME checks, ClamAV, hashing and no macro execution.

SUPPLY CHAIN

Signed images, SBOM, vulnerability scans and offline checksums.

AUDIT

Append-only decisions, overrides, AI outputs and integration activity.

RECOVERY

pgBackRest, object replication, tested restore and disaster runbook.

High-risk finance actions

Supplier bank changes, tolerance overrides, payment-readiness overrides and approval-policy changes require step-up authentication, segregation of duties and immutable evidence.

ERP and finance integration patterns

The connector layer translates external formats into canonical records. Customer-specific connector failures remain isolated from the core application.

Pattern	Best use	Control
REST / SOAP API	Modern ERP and finance applications	Idempotency key and signed webhook
SFTP files	Legacy batch imports and exports	Checksum, sequence and reconciliation
Read-only database	Approved legacy reporting access	No direct writes; change cursor
On-prem connector agent	ERP accessible only inside private network	Outbound TLS connection
Excel exchange	Manual or transitional operations	Mapping profile and approval

Synchronization ownership

Object	System of record	Direction
Supplier and item master	ERP or agreed master-data system	ERP → vProcly
Purchase request and sourcing evidence	vProcly	vProcly internal
Approved purchase order	ERP or configurable	Bi-directional with ownership rule
Receipt and invoice status	ERP	ERP → vProcly
Exception and resolution evidence	vProcly	vProcly → ERP reference
Payment transaction	Finance/ERP/bank	ERP → vProcly status

Reconciliation

Every connector needs checkpoints, external identifiers, payload hashes, retry state and a scheduled reconciliation job. Successful API calls alone do not prove that both systems agree.

Implementation roadmap

1

PHASE 1

Foundation

Tenancy, identity, PostgreSQL, MinIO, audit, event outbox, Kubernetes packaging.

6-8 weeks

2

PHASE 2

Excel + procurement

Mapping studio, imports, requests, approvals, RFQs, quotes and comparison.

10-14 weeks

3

PHASE 3

ExceptionOS

Detectors, scoring, assignment, escalation, playbooks and resolution evidence.

8-10 weeks

4

PHASE 4

Finance control

Budgets, reservations, commitments, invoice matching and payment readiness.

10-14 weeks

5

PHASE 5

Enterprise scale

HA packaging, connector SDK, analytics, local AI and air-gapped releases.

Ongoing

Architecture and product success measures

Area	Measure	Initial target
Excel import	Valid rows processed	≥ 5,000 rows/minute per worker
Import quality	Rows requiring correction	Trend downward by saved profile
Availability	Core procurement API	99.9% for HA profile
Recovery	Database recovery point	≤ 15 minutes where configured
Workflow	Stuck workflow detection	Under 5 minutes
Exceptions	Detection-to-assignment	Under 1 minute for event rules
Matching	Straight-through eligible invoices	Measured by category and tenant
Security	Critical vulnerability remediation	Defined customer SLA
Audit	Transactions with complete provenance	100%
Integration	Unreconciled synchronization records	Zero beyond configured SLA

Business outcomes

Cycle time

Request-to-PO duration

Control

Maverick and over-budget spend

Reliability

On-time delivery and supplier commitment

Assurance

Invoice mismatch and duplicate rate

Cash

Committed and upcoming payment visibility

Productivity

Manual comparison and follow-up hours

Final architecture decisions

Decision	Rationale
Procurement remains the product entry point	It creates the operational evidence required for finance control.
Finance expands through control, not full accounting	Reduces scope and integrates with existing statutory systems.
Excel is a governed channel	Mapping, staging, validation, provenance and rollback make it safe.
PostgreSQL is the primary store	Strong relationships, transactions and financial integrity.
Modular monolith first	Fast delivery with boundaries that allow later extraction.
Workers scale independently	Excel, OCR, AI and integrations have different resource profiles.
Temporal manages long-running execution	Approvals and supplier activity may last days or months.
Open and portable infrastructure	Kubernetes, Helm, MinIO, Keycloak and open telemetry support on-prem.
AI is advisory	Rules and humans retain control of financial actions.
Every decision has evidence	Auditability is a product capability, not only a log.

Recommended product definition

vProcly is an on-prem-ready procurement and financial-control platform that accepts operational data from Excel, ERP, APIs and documents; manages procurement workflows; and uses ExceptionOS to prevent delay, leakage and financial risk.

Build principle

Keep transactions deterministic, workflows durable, imports reversible, integrations isolated, AI explainable and every business decision traceable to source evidence.

Technology blueprint

Layer	Recommended technology	Portable alternative
Web	Next.js + TypeScript	Any OCI-hosted React application
Core API	NestJS modular monolith	Java/.NET service with same contracts
Excel service	Python + FastAPI + openpyxl/calamine	Java Apache POI
Database	PostgreSQL + pgvector	Customer-managed PostgreSQL
Object storage	MinIO	Any S3-compatible platform
Events	NATS JetStream	RabbitMQ or customer Kafka
Workflow	Temporal self-hosted	Customer-approved workflow engine
Identity	Keycloak	Customer OIDC/SAML provider
Cache	Redis	Valkey-compatible service
Search	PostgreSQL then OpenSearch	Customer search platform
Analytics	Materialized views then ClickHouse	Existing BI warehouse
AI inference	vLLM / provider gateway	Approved enterprise model endpoint
Deployment	Kubernetes + Helm + Argo CD	RKE2, OpenShift, private cloud
Observability	OpenTelemetry + Grafana stack	Customer observability platform

Avoid premature complexity

Kafka, service mesh, graph database, dedicated vector database, micro-frontends and dozens of microservices should be introduced only when a measured requirement justifies their operational cost.